

# SPEAR-3D: From Streaming Reconstruction to Explicit Reasoning—A Neuro-Symbolic Approach for 3D Visual Grounding

First A. Author, *Member, IEEE*, Second B. Author, and Third C. Author

**Abstract**—Open-vocabulary 3D scene instance reconstruction is a crucial capability for autonomous robots working in complex environments. It requires both efficient streaming 3D instance construction and accurate language-driven object localization within the constructed scene. Existing methods face two major challenges. First, online systems often divide 2D perception, semantic feature extraction, and 3D reconstruction into separate modules, leading to high latency and inconsistent scene representations. Second, directly using Large Language Models (LLMs) to spatial reasoning tasks may lead to spatial hallucinations, as the reasoning process lacks explicit constraints from reconstructed 3D geometry. To address these issues, we propose SPEAR-3D, a framework that combines prompt-conditioned Open-vocabulary streaming 3D scene instance reconstruction with explicit neuro-symbolic spatial reasoning. Our reconstruction module incrementally constructs an instance-aware 3D scene from RGB-D streams, and this streaming reconstruction stage operates at 10 FPS on an RTX 3090. Based on the reconstructed scene, we introduce a Neuro-Symbolic Query Graph Grounding (NS-Grounding) module that parses natural-language instructions into structured query graphs and matches them against a geometry-derived 3D scene graph. Experimental results show that SPEAR-3D achieves 10 FPS in streaming reconstruction, reaches 60.4 AP25 on ScanNet200, and improves grounding accuracy on NR3D/SR3D, validating the effectiveness of prompt-conditioned reconstruction and explicit geometric reasoning.

**Index Terms**—streaming 3D reconstruction, language-guided 3D object grounding, neuro-symbolic query graph grounding, spatial scene graph reasoning.

## I. INTRODUCTION

**R**EAL-TIME 3D scene understanding is a cornerstone for autonomous robots, enabling them to perceive, reconstruct, reason about, and interact with complex unstructured environments. Unlike traditional closed-set perception, which is limited to fixed training categories, open-vocabulary systems are designed to leverage natural-language category names or task-specific textual prompts during the inference phase. This capability is essential for embodied AI tasks such as mobile manipulation and semantic navigation [1], [2], where a robot needs to identify and locate objects from human descriptions.

Despite recent progress in 3D vision-language models, existing online systems still face a difficult trade-off between reconstruction efficiency and semantic generalization. Closed-set online systems such as Embodied-SAM [3] achieve real-time 3D instance construction, yet they are constrained by the semantic categories encountered during training. They

perform well on their training domains, such as ScanNet200 [4], but their performance may degrade when deployed in unseen environments such as SceneNN [5]. In contrast, open-vocabulary systems such as OnlineAnySeg [6] enable more flexible semantic recognition but typically rely on decoupled stages, including 2D segmentation, cross-frame association, and feature extraction. Such separation increases computational redundancy and limits the frame rate of online reconstruction.

To address these challenges, we propose SPEAR-3D, a framework for prompt-conditioned streaming 3D scene instance reconstruction and language-guided grounding. As shown in Fig. 1, Our first component is a **Category-Guided 3D Streaming Reconstruction** module. It leverages category prompts, SAM3 [7], and CLIP-style text embeddings to incrementally construct an instance-aware 3D scene from RGB-D streams. The reconstruction module is designed for streaming scene construction and achieves 10 FPS on an RTX 3090. We explicitly distinguish this reconstruction throughput from the latency of language-query processing. and this design enables semantic-driven 3D scene instance reconstruction and produces an explicit instance-aware 3D scene representation, providing a foundation for subsequent object retrieval, localization, and language-guided grounding.

Based on the reconstructed 3D instance scene, we introduce a **Neuro-Symbolic Query Graph Grounding (NS-Grounding)** module. Instead of treating the LLM as a black-box spatial reasoner, we use the LLM to parse natural-language descriptions into structured query graphs. These query graphs are then explicitly matched to a geometry-derived 3D scene graph constructed from the output of the streaming reconstruction process. In this way, language grounding is performed based on the reconstructed scene state, allowing object localization to be constrained by explicit 3D geometry and spatial relations.

It is important to clarify the scope of our open-vocabulary setting. SPEAR-3D performs prompt-conditioned open-vocabulary reconstruction that textual category vocabulary can be supplied or modified at inference time without the need for retraining. This differs from unconditional open-world object discovery. Objects whose names are not included in the inference-time prompt set may not be reconstructed as semantic instances. This design choice prioritizes efficient, task-relevant streaming reconstruction.

The primary contributions of this work are summarized as follows:

- We propose a **category-guided streaming 3D reconstruction** framework that incrementally constructs prompt-conditioned open-vocabulary 3D instance scenes from RGB-D streams and achieving **10 FPS** during the reconstruction stage.

Manuscript received [Date]; revised [Date]; accepted [Date]. Date of publication [Date]; date of current version [Date]. This letter was recommended for publication by Associate Editor [Name] and Editor [Name] upon evaluation of the reviewers' comments. (Corresponding author: [Name].)

The authors are with the [Your Department], [Your University/Institute], [City, Zip Code, Country] (e-mail: email1@domain.com; email2@domain.com).

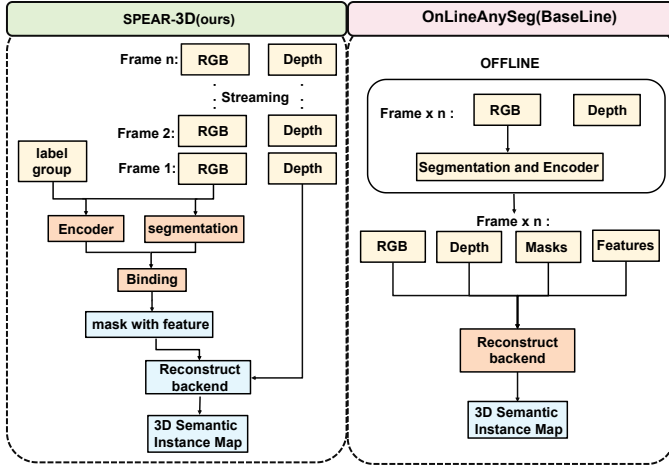


Fig. 1. Comparison between the baseline decoupled pipeline and SPEAR-3D. The baseline performs segmentation, feature encoding, and reconstruction in separate stages, whereas SPEAR-3D enables category-guided streaming 3D instance reconstruction from RGB-D inputs and produces an explicit instance-aware scene representation for downstream retrieval, localization, and language-guided grounding.

- We introduce **Neuro-Symbolic Query Graph Grounding (NS-Grounding)**, a method that grounds natural-language descriptions by matching LLM-parsed query graphs with scene graphs derived from the streaming reconstruction output.
- We comprehensively evaluate our framework on ScanNet200, SceneNN, NR3D, and SR3D datasets under a realistic streaming reconstructed setting, demonstrating that while maintaining a real-time speed of 10 FPS, SPEAR-3D achieves a peak  $AP_{25}$  of 60.4% in ScanNet200 and grounding accuracy achieves  $Acc@0.5$  65.9% in target object localization.

## II. RELATED WORK

### A. Real-Time 3D Instance Reconstruction

The ability to incrementally reconstruct and parse 3D scenes is a fundamental prerequisite for embodied agents. Early semantic SLAM and 3D tracking methods [8], [9] focused mainly on closed-set categories, limiting a robot’s ability to operate with novel object names. Subsequent works explored open-set multimodal scene construction [10], [11], but often lacked real-time instance-level fusion. Recent online systems accelerate 3D instance construction, but many of them either depend on closed-set semantics or introduce high latency through decoupled processing stages. SPEAR-3D addresses this trade-off by using prompt-conditioned category guidance for efficient streaming 3D instance reconstruction.

### B. Open-Vocabulary Segmentation and Foundation Models

To overcome closed-set limitations, researchers have integrated Vision-Language Models (VLMs) and visual foundation models into 3D perception. Offline methods such as MaskClustering [12] leverage CLIP [13] features to group open-vocabulary instances, but they are not designed for streaming reconstruction. OnlineAnySeg uses a decoupled pipeline with 2D

instance segmentation, cross-frame association, and semantic feature extraction. By incorporating open-vocabulary semantic features for cross-frame 3D instance fusion, this architecture enables explicit 3D scene instance reconstruction; however, the decoupled pipeline constrains reconstruction throughput and prevents streaming reconstruction.

As shown in Fig. 3, Our method uses SAM3 as a prompt-conditioned visual segmentation backbone integrated into a streaming 3D scene instance reconstruction process. The goal is not unconditional discovery of every object in an open world, but efficient reconstruction of task-relevant semantic instances specified by an inference-time prompt vocabulary. This method is practical for robotic tasks where object categories can be provided through task context, user instructions, environment priors, or benchmark inventories.

### C. 3D Visual Grounding and LLM-based Spatial Reasoning

3D visual grounding requires an agent to localize objects from free-form natural-language descriptions [14], [15]. Many recent pipelines attempt to inject 3D awareness into LLMs for spatial reasoning [16], [17]. While LLMs possess robust language parsing ability, directly employing them as black-box spatial reasoners may lead to spatial hallucinations. Inspired by visual programming and neuro-symbolic execution [18], [19], NS-Grounding separates linguistic parsing from geometric reasoning. The LLM produces a structured query graph, while the final spatial decision is made through explicit matching it against the reconstructed 3D scene graph.

## III. METHODOLOGY

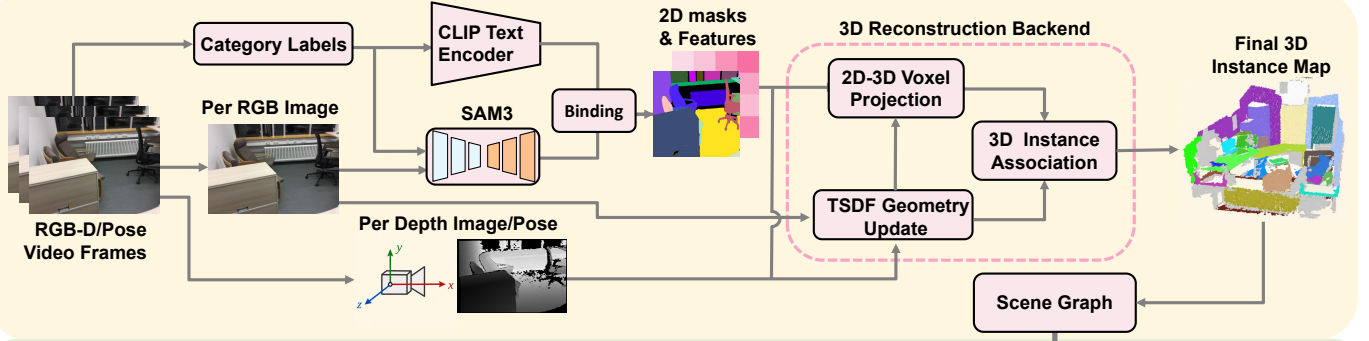
SPEAR-3D consists of two connected stages. (1) a *category-guided 3D streaming reconstruction* module that incrementally constructs an instance-aware 3D scene from RGB-D observations, and (2) a *neuro-symbolic query graph grounding* module that resolves natural-language queries against the reconstructed 3D scene. In this paper, *perception* is defined as to 2D visual mask generation and semantic feature extraction, while *3D reconstruction* or *scene construction* refers to the incremental fusion of RGB-D observations into a geometric and instance-level 3D representation.

### A. Category-Guided 3D Streaming Reconstruction

As shown in Fig. 2, the reconstruction module constructs a prompt-conditioned 3D instance scene using streaming RGB-D frames. Unlike conventional segment-then-classify pipelines, our method integrates semantic category information prior to 3D instance fusion, reducing per-mask recognition overhead and enabling efficient streaming updates.

a) *Prompt-Conditioned Open-Vocabulary Setting.*: At time step  $t$ , the prompt vocabulary is denoted as  $\mathcal{L}_t = \{l_k\}_{k=1}^{K_t}$ . The vocabulary can be provided at inference time from a task-level object list, user query context, dataset-level category inventory, or an external vocabulary source. Since  $\mathcal{L}_t$  is not fixed during model training, the system supports prompt-conditioned open-vocabulary 3D scene instance reconstruction. However, this setting should be distinguished from class-agnostic open-world discovery, the reconstruction recall is

## Category-Guided 3D Streaming Reconstruction



## Neuro-Symbolic Query Graph Grounding

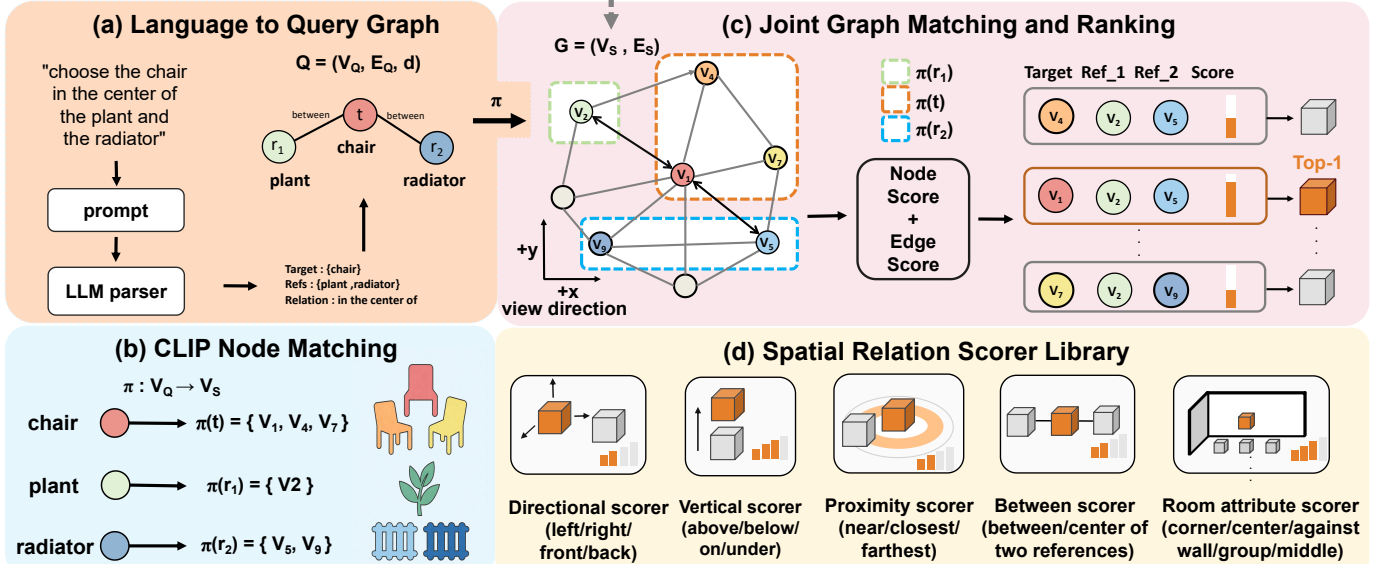


Fig. 2. Overall architecture of SPEAR-3D. The upper stage performs category-guided streaming 3D scene instance reconstruction from RGB-D sequences. Given the inference-time prompt vocabulary  $\mathcal{L}_t$ , SAM3 generates category-conditioned masks and CLIP provides label-bound semantic features, enabling prompt-conditioned 3D instance updates at 10 FPS. The reconstructed instance scene is converted into a spatial scene graph. The lower stage performs NS-Grounding on this reconstructed scene graph by parsing language into a query graph, selecting semantic candidates, and ranking target instances with explicit geometric relation scorers.

conditioned on the supplied prompts, and objects absent from  $\mathcal{L}_t$  may not be reconstructed as semantic instances.

*b) Parallel Semantic and Perceptual Streams.:* At time step  $t$ , given an RGB image  $I_t$ , a depth map  $D_t$ , and camera pose  $T_t$ , each category label  $l_k \in \mathcal{L}_t$  is processed through two streams. In the semantic stream, the category label is encoded by a CLIP text encoder  $\Psi(\cdot)$ :

$$\mathbf{e}_k = \frac{\Psi(l_k)}{\|\Psi(l_k)\|}. \quad (1)$$

In the perceptual stream, the same prompt vocabulary is provided to SAM3:

$$\{M_{t,k}\}_{k=1}^{K_t} = \text{SAM3}(I_t, \mathcal{L}_t), \quad (2)$$

where  $M_{t,k}$  denotes the 2D mask associated with category prompt  $l_k$  at frame  $t$ .

*c) Label-Mediated Feature Binding.:* Semantic identity is assigned before cross-frame 3D association. Instead of extracting a separate text feature for each cropped mask,

we bind the precomputed category embedding  $\mathbf{e}_k$  to its corresponding mask:

$$\mathbf{f}_{t,k} \xleftarrow{\text{ID}(l_k)} \mathbf{e}_k. \quad (3)$$

$\mathbf{f}_{t,k}$  is the category embedding corresponding to the mask after binding. This yields a compact instance observation  $(M_{t,k}, \mathbf{f}_{t,k}, l_k)$  and reduces ambiguity in the subsequent 3D fusion stage.

*d) 3D Scene Instance Reconstruction Backend:* After obtaining category-conditioned 2D masks and label-bound semantic features, we adopt the online reconstruction backend of OnlineAnySeg for geometric lifting and temporal association. Specially, each RGB-D frame is first integrated into a 3D voxel-hashed TSDF geometry state  $\mathcal{G}_t$ :

$$\mathcal{G}_t = \mathcal{U}(\mathcal{G}_{t-1}, I_t, D_t, T_t), \quad (4)$$

and each 2D mask is projected into the active 3D geometry  $\mathcal{V}_{t,k}$ :

$$\mathcal{V}_{t,k} = \Pi(M_{t,k}, D_t, T_t; \mathcal{G}_t). \quad (5)$$

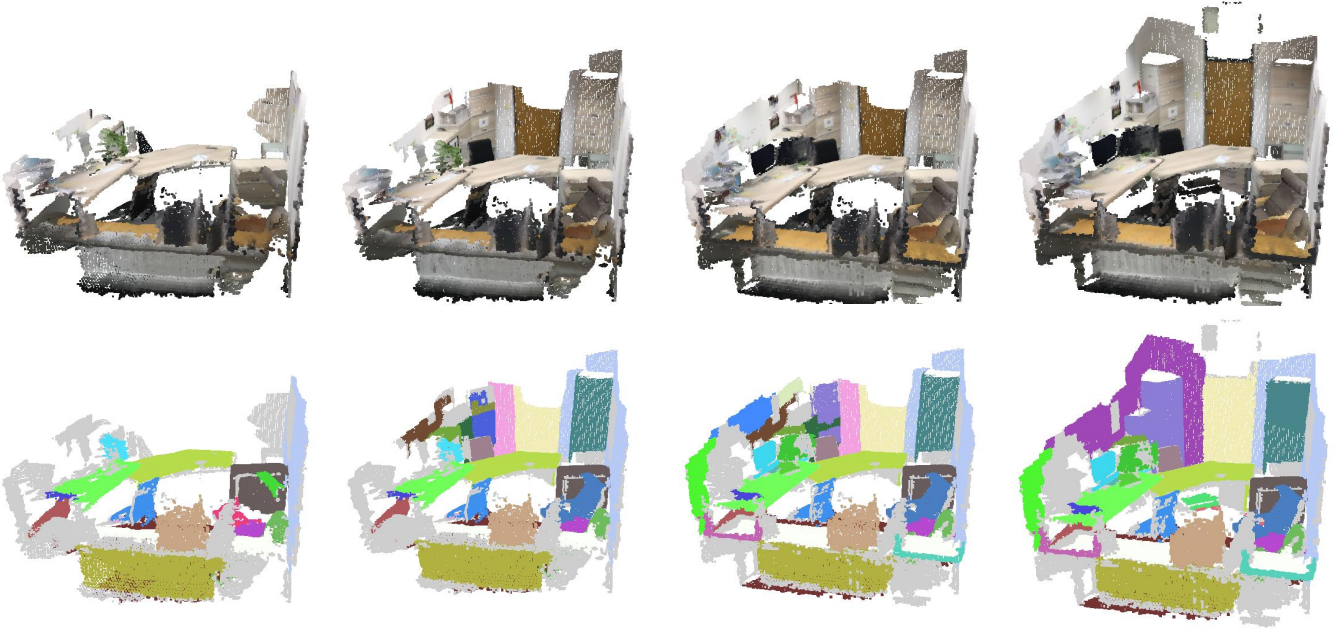


Fig. 3. SAM3-driven streaming 3D reconstruction. The top row illustrates incremental geometric reconstruction from RGB-D sequences. The bottom row visualizes instance-level semantic masks generated by SAM3, showing how object observations are fused across viewpoints to form a coherent 3D instance library.

For temporal association, the backend computes an overlap-, semantic-, and geometry-aware matching score  $S_{kj}$ :

$$S_{kj} = \lambda_s \mathcal{A}(\mathbf{f}_{t,k}, \mathbf{f}_j) + \lambda_g S_{kj}^{\text{geo}} + \lambda_o R_{kj}, \quad (6)$$

where  $\mathcal{A}(\mathbf{f}_{t,k}, \mathbf{f}_j)$  is the cosine similarity between the semantic embeddings of the current observation and the associated reconstructed instance,  $S_{kj}^{\text{geo}}$  is the geometric similarity computed from FCGF descriptors extracted from the corresponding voxelized observations, and  $R_{kj}$  is the 3D overlap ratio. Parameters  $\lambda_s$ ,  $\lambda_g$  and  $\lambda_o$  are set the same as in OnlineAnySeg. The resulting voxelized observations and matching scores are then passed to our category-guided streaming instance update module.

*e) Category-Guided Streaming Instance Update.:* Given voxelized observations  $\{(V_{t,k}, \mathbf{f}_{t,k}, l_k)\}_{k=1}^{K_t}$  and matching score matrix  $\mathbf{S}_t = [S_{kj}]$ , we update:

$$\mathcal{I}_t = \mathcal{R}(\mathcal{I}_{t-1}, \{(V_{t,k}, \mathbf{f}_{t,k}, l_k)\}_{k=1}^{K_t}, \mathbf{S}_t). \quad (7)$$

The updated library  $\mathcal{I}_t$  is a prompt-conditioned 3D instance scene library used to construct the scene graph required by NS-Grounding. The streaming reconstruction stage runs at approximately 10 FPS on an RTX 3090.

### B. NS-Grounding: Neuro-Symbolic Query Graph Grounding

NS-Grounding performs natural-language object grounding over the reconstructed 3D instance scene. We formulate grounding as matching a language-derived structured query graph to a geometry-derived scene graph [20], [21]. From the 3D scene instance library  $\mathcal{I}_t$ , we construct a spatial scene graph  $G = (V_S, E_S)$ , where  $V_S$  nodes represent reconstructed object instances with category, position, and size, and  $E_S$  edges encode spatial relations.

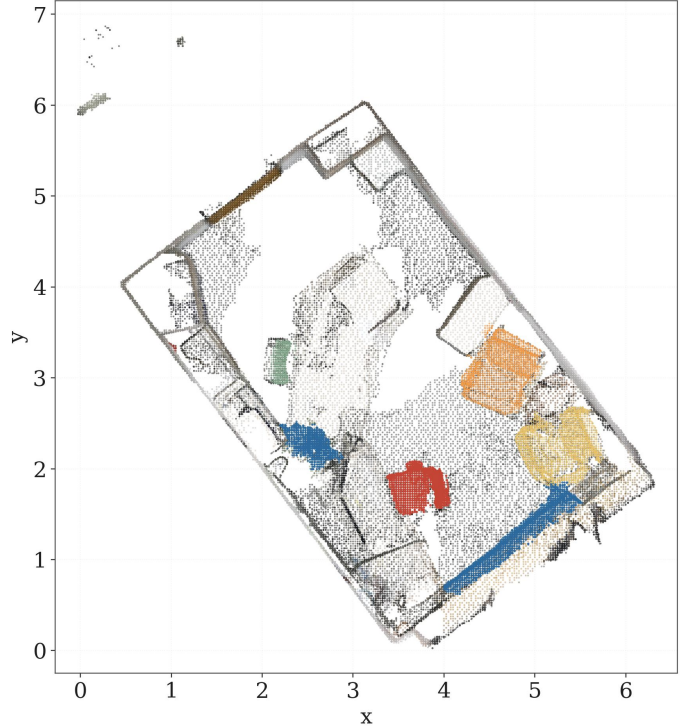


Fig. 4. 2D bird's-eye view of topological grounding. The target object is localized in the XY-plane by resolving the *between* constraint relative to reference anchors while effectively rejecting distractors.

*1) Language to Query Graph:* As shown in Fig 2(a), An LLM parses the input expression into a structured query graph  $Q = (V_Q, E_Q, d)$ , where  $V_Q$  represents target and reference nodes,  $E_Q$  contains typed relational edges, and  $d$  represents

an optional viewer direction.

2) *CLIP Node Matching*: As shown in Fig 2(b), To align target query labels with reconstructed object categories, we compute CLIP similarity between query label  $\ell_u$  and scene class  $c$ :

$$s_{\text{clip}}(\ell_u, c) = \cos(f_{\text{text}}(\ell_u), f_{\text{text}}(c)). \quad (8)$$

Top-matched reconstructed instances form candidate sets for each query node.

3) *Joint Graph Matching and Ranking*: Grounding is formulated as a structured assignment problem. As shown in Fig 2(c), For an injective mapping  $\pi : V_Q \rightarrow V_S$  and direction  $d$ , the compatibility score is:

$$S(\pi | d) = \sum_{u \in V_Q} \lambda_u s_{\text{node}}(u, \pi(u) | d) + \sum_{e \in E_Q} \lambda_e s_{\text{rel}}(e, \pi | d), \quad (9)$$

where  $u \in V_Q$  denotes a query node and  $e \in E_Q$  denotes a relation edge connecting two query nodes.  $s_{\text{node}}$  evaluates node-level semantic consistency, while  $s_{\text{rel}}$  measures the satisfaction of the spatial relation encoded by edge  $e$ .  $\lambda_u$  and  $\lambda_e$  are the weights for node matching and relation matching, respectively, with  $\lambda_u + \lambda_e = 1$ . The influence of these parameters is analyzed in the experimental section.

4) *Spatial Relation Scorer Library*: As shown in Fig 2(d), we compute  $s_{\text{rel}}$  through a geometric scorer library. Let  $o_u = \pi(u)$  and  $o_v = \pi(v)$  be reconstructed scene instances with centers  $p_{o_u}$  and  $p_{o_v}$ . All distances are normalized by scene-adaptive scales. We use  $\sigma(x) = (1 + e^{-x})^{-1}$  and  $\phi(x) = e^{-x}$ . In the following,  $s_{\text{dir}}$ ,  $s_{\text{near}}$ ,  $s_{\text{rank}}$ ,  $s_{\text{vert}}$ ,  $s_{\text{contact}}$ ,  $s_{\text{side}}$ ,  $s_{\text{between}}$  denote the scores of various correspondence relationships.

a) *Directional Relations*.: For directional relations, including left, right, front, and behind, we project displacement  $p_{o_u} - p_{o_v}$  onto the queried axis  $\delta_{\parallel}$  and the orthogonal axis  $\delta_{\perp}$ :

$$s_{\text{dir}}(o_u, o_v | d) = \sigma(\delta_{\parallel}) \phi(|\delta_{\perp}|). \quad (10)$$

b) *Proximity and Comparative Distance*.: Proximity uses horizontal distance  $d_{xy}(o_u, o_v) = \|p_{o_u}^{xy} - p_{o_v}^{xy}\|_2$  and an XY overlap bonus  $\gamma_{xy}(o_u, o_v)$ :

$$s_{\text{near}}(o_u, o_v) = \phi(d_{xy}(o_u, o_v)) \gamma_{xy}(o_u, o_v). \quad (11)$$

Comparative relations evaluate instance  $o_u$  relative to its candidate set  $\mathcal{C}(u)$  using the 3D distance  $r(o, o_v) = \|p_o - p_{o_v}\|_2$  and the normalized rank  $\rho(o_u, o_v)$ :

$$s_{\text{rank}}(o_u, o_v) = \rho(o_u, o_v) + \phi(|r(o_u, o_v) - r^*|). \quad (12)$$

c) *Vertical and Contact Relations*.: Vertical relations combine height ordering with horizontal alignment  $a_{xy}(o_u, o_v)$ :

$$s_{\text{vert}}(o_u, o_v) = \sigma(p_{o_u}^z - p_{o_v}^z) a_{xy}(o_u, o_v). \quad (13)$$

Contact-sensitive relations add a vertical contact term  $\phi(g_z(o_u, o_v))$  based on the surface gap  $g_z$ :

$$s_{\text{contact}}(o_u, o_v) = s_{\text{vert}}(o_u, o_v) \phi(g_z(o_u, o_v)). \quad (14)$$

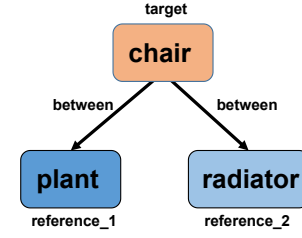


Fig. 5. Structured query graph parsing. The instruction separates the target entity (*chair*) from contextual anchors (*plant*, *radiator*) through an explicit ternary geometric constraint (*between*).

d) *Room-side and Between Relations*.: For the *same-side-of-room* relation, we measure each object's affinity to the four room boundaries  $\mathcal{W} = \{x_{\min}, x_{\max}, y_{\min}, y_{\max}\}$ . Given the distance  $\delta_w(o)$  from object  $o$  to wall  $w$ , the wall affinity is

$$\alpha_w(o) = \exp\left(-\frac{\delta_w(o)}{\tau_{\text{wall}}}\right). \quad (15)$$

The room-side relation score is then defined as the best shared wall affinity:

$$s_{\text{side}}(o_u, o_v) = \max_{w \in \mathcal{W}} \min(\alpha_w(o_u), \alpha_w(o_v)). \quad (16)$$

For the ternary *between* relation, let  $c_u, c_{v_1}, c_{v_2} \in \mathbb{R}^3$  denote object centers and  $m = (c_{v_1} + c_{v_2})/2$  the midpoint of the two anchors. We combine midpoint proximity, distance balance, and in-plane alignment:

$$s_{\text{mid}} = \exp\left(-\frac{\|c_u - m\|_2}{\tau_{\text{mid}}}\right), \quad (17)$$

$$s_{\text{bal}} = \exp\left(-\frac{\|c_u - c_{v_1}\|_2 - \|c_u - c_{v_2}\|_2}{\tau_{\text{mid}}}\right), \quad (18)$$

$$s_{\text{align}} = \exp\left(-\frac{d_{xy}(c_u, \overline{c_{v_1} c_{v_2}})}{\tau_{\text{align}}}\right), \quad (19)$$

and compute

$$s_{\text{between}}(o_u; o_{v_1}, o_{v_2}) = s_{\text{mid}} s_{\text{bal}} s_{\text{align}}. \quad (20)$$

If the viewer direction is unknown, final grounding optimizes over canonical directions:

$$S(\pi) = \max_{d \in \{+x, -x, +y, -y\}} S(\pi | d), \quad \hat{o} = \hat{\pi}(t). \quad (21)$$

## IV. EXPERIMENTS

### A. Experimental Settings

We evaluate SPEAR-3D from two complementary perspectives: streaming 3D instance reconstruction and language grounding over reconstructed scenes. For reconstruction, we use ScanNet200 and SceneNN, and report AP, AP50, and AP25. For grounding, we use NR3D and SR3D, and report Mean Distance (MeanDist) as well as Acc@0.3 and Acc@0.5, measured against ground-truth target centroids. Following the prompt-conditioned setting of our method, the inference-time prompt vocabulary is set to the benchmark category inventory of each dataset, which makes the evaluation protocol explicit and reproducible.

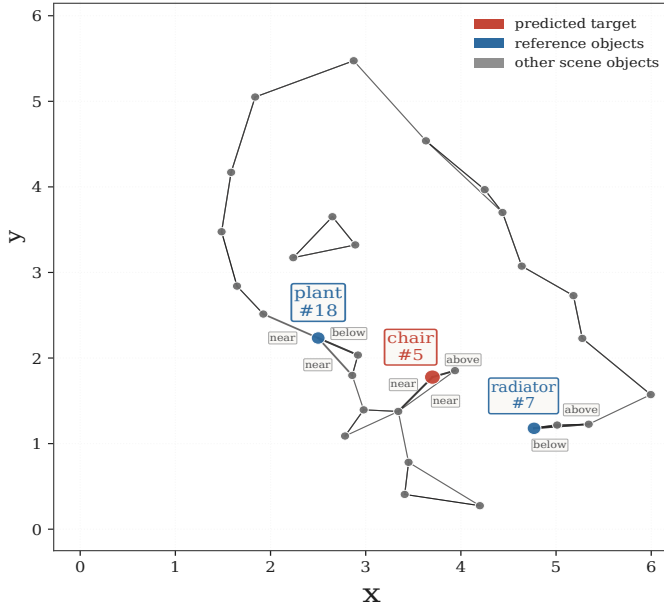


Fig. 6. 3D scene graph topology. The graph encodes explicit spatial relations as the substrate for structural graph matching.

All experiments are conducted on an RTX 3090 GPU. The streaming reconstruction stage runs at approximately 10 FPS, while NS-Grounding is invoked on demand and takes about 2 seconds per query using the Qwen3-plus API, including query-graph parsing and explicit graph matching. We report these two runtimes separately, since reconstruction throughput should not be confused with language-query latency.

### B. Hyperparameter Settings.

In the joint graph matching module, the node-relation weighting parameters  $\lambda_u$  and  $\lambda_e$  control the trade-off between node matching and relation matching, respectively, with  $\lambda_u + \lambda_e = 1$ . We sweep  $\lambda_u \in \{0.25, 0.50, 0.75\}$  on NR3D using Qwen3-plus to study the influence of this balance. As shown in Table III The best performance is obtained when  $\lambda_u = 0.50$ , which indicates that semantic consistency and spatial relation reasoning should contribute equally in our grounding setting.

### C. Quantitative Analysis

*a) Reconstruction Performance.*: As shown in Table I, SPEAR-3D balances reconstruction accuracy, semantic flexibility, and computational efficiency. Compared with the decoupled OnlineAnySeg pipeline, our SAM3-integrated streaming reconstruction stage achieves **10 FPS**, while OnlineAnySeg operates at merely 4 FPS when executing streaming reconstruction for 3D instance scenes. On ScanNet200, SPEAR-3D achieves an  $AP_{25}$  of 60.4%, outperforming OnlineAnySeg by 6.9 percentage points. The performance drop on SceneNN is primarily attributed to the missing annotations of small or ambiguous objects in the dataset, which conflicts with our category-driven reconstruction strategy. Unlike ScanNet200, SceneNN’s incomplete semantic inventory means objects omitted from the task-relevant prompt vocabulary are skipped during the

streaming stage, thereby impacting strict AP metrics. Ultimately, this result reflects a calculated trade-off where SPEAR-3D prioritizes computational efficiency and task-driven precision over exhaustive scene recall. Additionally, while ESAM exhibits higher quantitative metrics due to its supervised training natively optimized on the ScanNet dataset, it operates as a closed-set validator and lacks the generalized open-vocabulary capability intrinsic to our framework.”

*b) Grounding Performance.*: As shown in Table II, NS-Grounding consistently improves grounding performance across different LLM backbones. Under the same reconstructed-scene input, explicit graph matching outperforms direct black-box LLM reasoning, indicating that structured geometric constraints help reduce spatial ambiguity and improve target localization. The best result is achieved by Qwen3-plus with NS-Grounding, which obtains a MeanDist of 0.96 m and Acc@0.3/Acc@0.5 of 64.7%/65.9% on SR3D, demonstrating the effectiveness of combining explicit scene-graph reasoning with modern LLMs.

### D. Ablation Studies

*a) Ablation of NS-Grounding.*: Table II isolates the contribution of NS-Grounding by comparing each LLM backbone with and without the proposed graph-based reasoning module. Across all backbones, adding NS-Grounding consistently improves both distance-based and threshold-based grounding metrics on NR3D and SR3D, indicating that the module is beneficial regardless of the underlying language model. The gains are particularly clear for models that rely more heavily on direct prompting. For instance, Qwen3-plus and GLM-5 show large improvements after enabling NS-Grounding, while deepseek-v3.2 also benefits consistently, albeit from a stronger baseline. On SR3D, all three backbones achieve substantially better localization accuracy and lower MeanDist once explicit graph matching is introduced. These results demonstrate that the proposed module contributes a stable and model-agnostic improvement by replacing implicit LLM guessing with structured scene-graph reasoning.

*b) Sensitivity to Node-Relation Weighting.*: Table III evaluates the influence of the node-relation weighting parameter  $\lambda_u$ . The best performance is achieved at  $\lambda_u = 0.50$ , suggesting that semantic consistency and spatial relation reasoning should be balanced in our grounding framework. More importantly, the results remain stable across a relatively wide range of settings. When  $\lambda_u$  varies from 0.25 to 0.75, the performance changes only moderately, and all configurations produce comparable grounding accuracy. This indicates that the proposed method is not sensitive to the exact choice of  $\lambda_u$ , which reduces the need for careful hyperparameter tuning and makes the framework more practical for deployment across different datasets and LLM backbones.

### E. Qualitative Analysis and Reasoning Logic

As shown in Fig. 5. To illustrate why NS-Grounding mitigates spatial hallucinations, we visualize the reasoning trajectory for the instruction: “choose the chair that is in the center of the plant and the radiator”.

TABLE I  
FULL-SEQUENCE 3D SCENE INSTANCE RECONSTRUCTION RESULTS ON SCANNET200 AND SCENENNN.

| Method              | Streaming | Online | Zero-shot | ScanNet200  |                  |                  | SceneNN     |                  |                  | FPS       |
|---------------------|-----------|--------|-----------|-------------|------------------|------------------|-------------|------------------|------------------|-----------|
|                     |           |        |           | AP          | AP <sub>50</sub> | AP <sub>25</sub> | AP          | AP <sub>50</sub> | AP <sub>25</sub> |           |
| EmbodiedSAM [3]     | ✓         | ✓      | ×         | <b>28.8</b> | <b>42.7</b>      | <u>54.2</u>      | <b>20.1</b> | <u>32.5</u>      | 46.3             | <b>10</b> |
| OVIR-3D [22]        | ×         | ×      | ✓         | 14.4        | 27.5             | 38.8             | 12.3        | 24.4             | 34.6             | -         |
| MaskClustering [12] | ×         | ×      | ✓         | 19.7        | 36.4             | 51.4             | 16.3        | 31.7             | 46.2             | -         |
| SAM3D [23]*‡        | ×         | ×      | ✓         | 17.8        | 30.6             | 48.5             | -           | -                | -                | 8         |
| SAM3D [23]*†        | ✓         | ✓      | ✓         | 9.6         | 24.8             | 49.6             | 9.1         | 21.3             | 43.4             | 8         |
| OnlineAnySeg [6]    | ×         | ✓      | ✓         | 18.6        | 36.1             | 53.5             | <u>18.1</u> | <b>35.3</b>      | <b>59.5</b>      | 4         |
| <b>Ours</b>         | ✓         | ✓      | ✓         | <u>22.6</u> | <u>40.4</u>      | <b>60.4</b>      | 15.6        | 30.4             | <u>54.7</u>      | <u>10</u> |

TABLE II  
ABLATION OF THE PROPOSED NS-GROUNDING MODULE ON NR3D AND SR3D. GROUNDING IS EVALUATED ON INSTANCES PRODUCED BY STREAMING RECONSTRUCTION, NOT ON COMPLETE GROUND-TRUTH SCANNET POINT CLOUDS.

| LLM               | NS-Grounding | NR3D        |             |             | SR3D        |             |             |
|-------------------|--------------|-------------|-------------|-------------|-------------|-------------|-------------|
|                   |              | MeanDist ↓  | Acc@0.3 ↑   | Acc@0.5 ↑   | MeanDist ↓  | Acc@0.3 ↑   | Acc@0.5 ↑   |
| Qwen3-plus        | ×            | 1.36        | 38.7        | 43.5        | 1.32        | 53.8        | 54.8        |
| <b>Qwen3-plus</b> | ✓            | <b>1.02</b> | <b>56.3</b> | <b>58.8</b> | <b>0.96</b> | <b>64.7</b> | <b>65.9</b> |
| deepseek-v3.2     | ×            | 1.33        | 42.0        | 44.6        | 1.24        | 53.1        | 54.1        |
| deepseek-v3.2     | ✓            | 1.27        | 51.1        | 54.3        | 1.00        | 64.8        | 66.0        |
| GLM-5 [24]        | ×            | 1.35        | 37.8        | 41.5        | 1.28        | 52.5        | 51.9        |
| GLM-5 [24]        | ✓            | 1.18        | 50.2        | 53.4        | 0.97        | 65.6        | 67.0        |

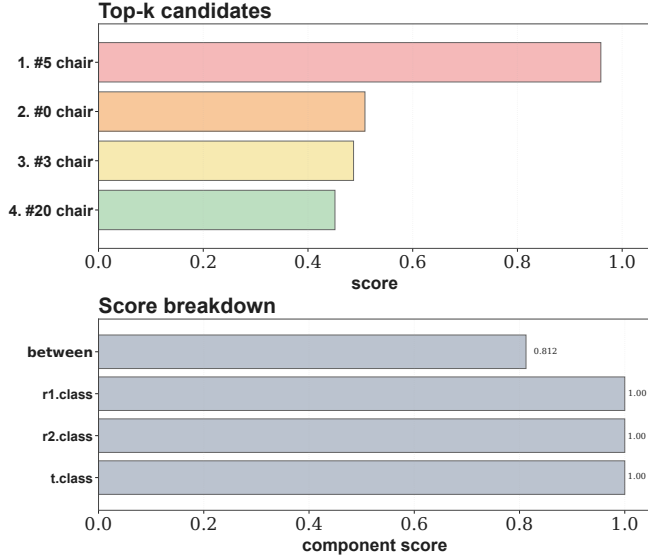


Fig. 7. Explicit score decomposition. Although multiple candidates have strong semantic alignment, *chair #5* is selected because it satisfies the dominant geometric relation.

First, the streaming reconstruction module incrementally constructs semantic 3D instances from RGB-D data. By binding semantic prompts before fusion, the target chair and reference objects are consolidated into stable reconstructed instances rather than fragmented frame-level observations.

Second, As shown in Fig. 6. These instances are converted

TABLE III  
EFFECT OF DIFFERENT  $\lambda_u$  ON NR3D USING QWEN3-PLUS.

| $\lambda_u$ | MeanDist ↓    | Acc@0.3 ↑   | Acc@0.5 ↑   |
|-------------|---------------|-------------|-------------|
| 0.25        | 1.2550        | 48.1        | 51.6        |
| <b>0.50</b> | <b>1.0200</b> | <b>56.3</b> | <b>58.8</b> |
| 0.75        | 1.0840        | 53.4        | 57.8        |

into a spatial scene graph  $G = (V_S, E_S)$ , turning raw reconstructed geometry into a relational abstraction that can be queried compositionally.

Third, the natural-language query is parsed into a structured graph and matched against the reconstructed scene graph. As shown in Fig. 7, multiple candidate chairs align semantically, but *chair #5* receives the highest geometric relation score for the *between* constraint. Distractor chairs fail the topological constraint, showing that semantic identity alone is insufficient for 3D disambiguation. In contrast, the black-box LLM baseline incorrectly selects distractor *chair #3*, as it fails to reliably evaluate the continuous *between* metric without explicit geometric verification. Finally, as shown in Fig. 4 and Fig. 8, the predicted target aligns with the ground-truth target location. This transparent reasoning process externalizes spatial reasoning into an inspectable geometric procedure.

## V. CONCLUSION

We presented SPEAR-3D, a neuro-symbolic framework that unifies prompt-conditioned streaming 3D instance reconstruc-

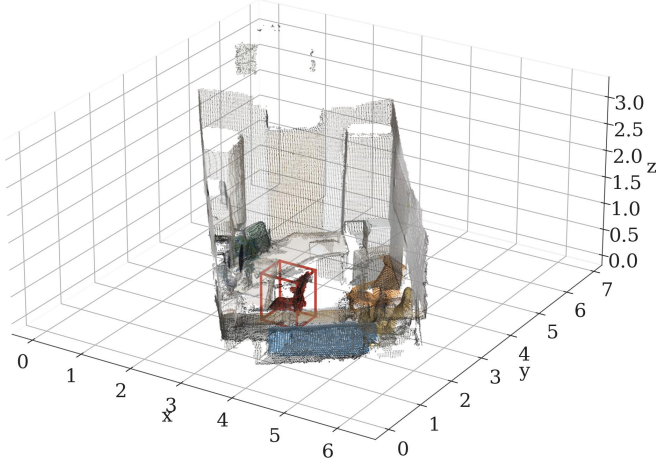


Fig. 8. Final 3D spatial localization. The red bounding box indicates the predicted target extent in the reconstructed scene.

tion with explicit language grounding. The main contribution of this work is to couple category-guided, instance-aware reconstruction with graph-based reasoning, enabling open-vocabulary 3D scene understanding to proceed in real time while remaining structurally interpretable. By binding semantic prompts before 3D fusion and then grounding language over reconstructed scene graphs, SPEAR-3D addresses two key challenges in existing pipelines: the latency introduced by decoupled online perception modules and the spatial hallucinations that may arise when large language models are used as black-box reasoners.

Experiments on ScanNet200, SceneNN, NR3D, and SR3D demonstrate that SPEAR-3D improves both reconstruction efficiency and grounding accuracy under the same streaming reconstructed-scene setting. These results suggest that explicit geometric reasoning can serve as a reliable bridge between streaming perception and natural-language grounding, offering a practical route toward more robust embodied scene understanding. More broadly, our framework shows that semantic flexibility and real-time performance do not have to be mutually exclusive, and that they can be jointly realized through a carefully designed reconstruction-and-reasoning pipeline.

A current limitation is that reconstruction recall is still conditioned on the supplied prompt vocabulary, so objects outside the prompt set may be missed when the category inventory is incomplete. In future work, we will investigate hybrid category-guided and class-agnostic proposal mechanisms, together with more efficient query processing and broader open-world reconstruction, to further improve recall, robustness, and scalability in streaming 3D perception.

## REFERENCES

- [1] A. Das, S. Datta, G. Gkioxari, S. Lee, D. Parikh, and D. Batra, “Embodied question answering,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 1–10.
- [2] D. S. Chaplot, D. P. Gandhi, A. Gupta, and R. R. Salakhutdinov, “Object goal navigation using goal-oriented semantic exploration,” *Advances in Neural Information Processing Systems*, vol. 33, pp. 4247–4258, 2020.
- [3] X. Xu, H. Chen, L. Zhao, Z. Wang, J. Zhou, and J. Lu, “Embodiedsam: Online segment any 3d thing in real time,” *arXiv preprint arXiv:2408.11811*, 2024.
- [4] D. Rozenberszki, O. Litany, and A. Dai, “Language-grounded indoor 3d semantic segmentation in the wild,” in *European conference on computer vision*. Springer, 2022, pp. 125–141.
- [5] B.-S. Hua, Q.-H. Pham, D. T. Nguyen, M.-K. Tran, L.-F. Yu, and S.-K. Yeung, “Scenenn: A scene meshes dataset with annotations,” in *2016 fourth international conference on 3D vision (3DV)*. Ieee, 2016, pp. 92–101.
- [6] Y. Tang, J. Zhang, Y. Lan, Y. Guo, D. Dong, C. Zhu, and K. Xu, “Onlineanyseg: Online zero-shot 3d segmentation by visual foundation model guided 2d mask merging,” in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 3676–3685.
- [7] N. Carion, L. Gustafson, Y.-T. Hu, S. Debnath, R. Hu, D. Suris, C. Ryali, K. V. Alwala, H. Khedr, A. Huang *et al.*, “Sam 3: Segment anything with concepts,” *arXiv preprint arXiv:2511.16719*, 2025.
- [8] J. McCormac, A. Handa, A. Davison, and S. Leutenegger, “Semanticfusion: Dense 3d semantic mapping with convolutional neural networks,” in *2017 IEEE International Conference on Robotics and automation (ICRA)*. IEEE, 2017, pp. 4628–4635.
- [9] M. Runz, M. Buffier, and L. Agapito, “Maskfusion: Real-time recognition, tracking and reconstruction of multiple moving objects,” in *2018 IEEE international symposium on mixed and augmented reality (ISMAR)*. IEEE, 2018, pp. 10–20.
- [10] K. M. Jatavallabhula, A. Kuwajerwala, Q. Gu, M. Omama, T. Chen, A. Maalouf, S. Li, G. Iyer, S. Saryazdi, N. Keetha *et al.*, “Conceptfusion: Open-set multimodal 3d mapping,” *arXiv preprint arXiv:2302.07241*, 2023.
- [11] S. Peng, K. Genova, C. Jiang, A. Tagliasacchi, M. Pollefeys, T. Funkhouser *et al.*, “Openscene: 3d scene understanding with open vocabularies,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2023, pp. 815–824.
- [12] M. Yan, J. Zhang, Y. Zhu, and H. Wang, “Maskclustering: View consensus based mask graph clustering for open-vocabulary 3d instance segmentation,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 28 274–28 284.
- [13] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark *et al.*, “Learning transferable visual models from natural language supervision,” in *International conference on machine learning*. PmlR, 2021, pp. 8748–8763.
- [14] D. Z. Chen, A. X. Chang, and M. Nießner, “Scanrefer: 3d object localization in rgb-d scans using natural language,” in *European conference on computer vision*. Springer, 2020, pp. 202–221.
- [15] Y. Zhang, Z. Gong, and A. X. Chang, “Multi3drefer: Grounding text description to multiple 3d objects,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 15 225–15 236.
- [16] Y. Hong, H. Zhen, P. Chen, S. Zheng, Y. Du, Z. Chen, and C. Gan, “3d-llm: Injecting the 3d world into large language models,” *Advances in Neural Information Processing Systems*, vol. 36, pp. 20 482–20 494, 2023.
- [17] C. Zhu, T. Wang, W. Zhang, J. Pang, and X. Liu, “Llava-3d: A simple yet effective pathway to empowering llms with 3d-awareness,” *arXiv preprint arXiv:2409.18125*, 2024.
- [18] D. Suris, S. Menon, and C. Vondrick, “Vipergpt: Visual inference via python execution for reasoning,” in *Proceedings of the IEEE/CVF international conference on computer vision*, 2023, pp. 11 888–11 898.
- [19] T. Gupta and A. Kembhavi, “Visual programming: Compositional visual reasoning without training,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2023, pp. 14 953–14 962.
- [20] J. Wald, H. Dhamo, N. Navab, and F. Tombari, “Learning 3d semantic scene graphs from 3d indoor reconstructions,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 3961–3970.
- [21] Q. Gu, A. Kuwajerwala, S. Morin, K. M. Jatavallabhula, B. Sen, A. Agarwal, C. Rivera, W. Paul, K. Ellis, R. Chellappa *et al.*, “Conceptgraphs: Open-vocabulary 3d scene graphs for perception and planning,” in *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2024, pp. 5021–5028.
- [22] S. Lu, H. Chang, E. P. Jing, A. Boularias, and K. Bekris, “Ovir-3d: Open-vocabulary 3d instance retrieval without training on 3d data,” in *Conference on Robot Learning*. PMLR, 2023, pp. 1610–1620.
- [23] Y. Yang, X. Wu, T. He, H. Zhao, and X. Liu, “Sam3d: Segment anything in 3d scenes,” *arXiv preprint arXiv:2306.03908*.
- [24] Z. Du, Y. Qian, X. Liu, M. Ding, J. Qiu, Z. Yang, and J. Tang, “Glm: General language model pretraining with autoregressive blank infilling,” in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2022, pp. 320–335.